CAPSTONE PROJECT PROPOSAL (draft)

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Topic Title: Food Delivery Optimization Dashboard using Machine Learning and Data Analytics

Project Overview

1. Problem Statement

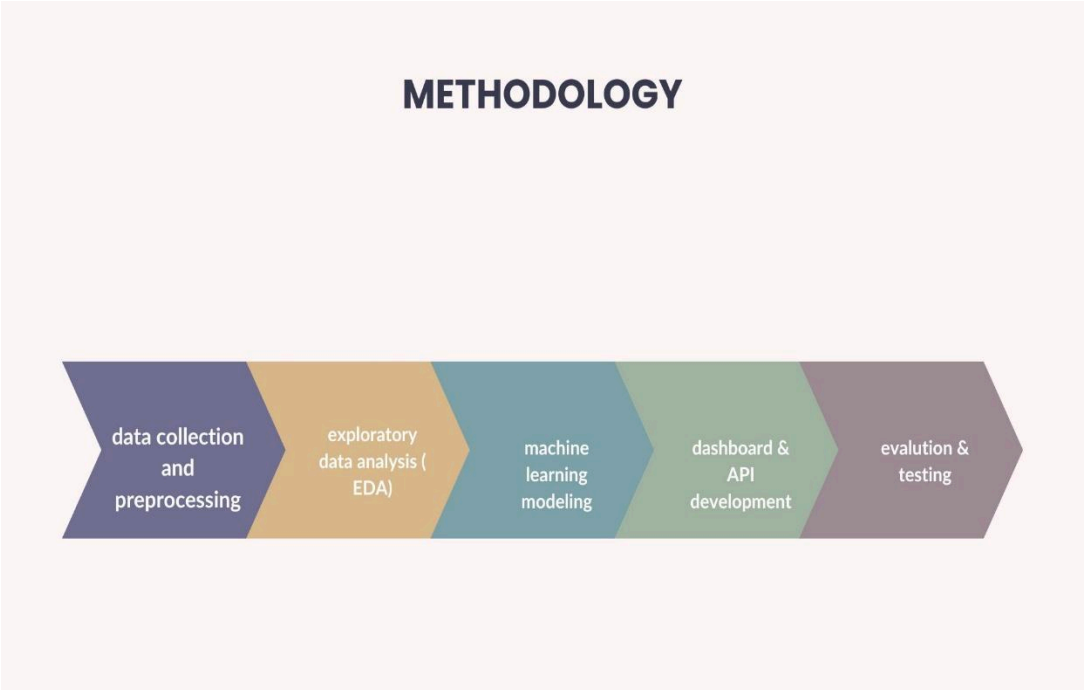
Food delivery companies face challenges in balancing delivery speed, customer satisfaction, and operational costs. Delays can result from factors such as driver availability, weather, traffic, or order volume — leading to customer dissatisfaction and inefficiency.

This project aims to develop a **data-driven dashboard** that identifies key factors affecting delivery performance and uses **machine learning models** to predict delivery times and optimize operational efficiency.

2. Objectives

- To analyze historical delivery data and identify patterns affecting delivery time and customer satisfaction.
- To build a **predictive model** that estimates delivery time based on input features (e.g., distance, order size, weather).
- To visualize key metrics and insights in a **Tableau** for quick decision-making.
- To design a REST API endpoint that provides real-time delivery time predictions for new orders.
- To generate actionable recommendations for improving performance and customer experience.

3.



4. Tools and Technologies

- **Programming:** Python, R
- **Libraries:** Pandas
- **Visualization:** Tableau
- **Backend/API:** Flask or FastAPI
- **Version Control:** GitHub
- **Project Management:** Jira

5. Timetable

No	Duration	Goal To Reach
1	Oct 17 – Oct 31	Topic approval, data sourcing & cleaning
2	Nov 1 – Nov 15	Exploratory Data Analysis & visualization
3	Nov 16 – Nov 30	Model building & evaluation
4	Dec 1 – Dec 15	Dashboard and API development
5	Dec 16 – Dec 31	Testing, documentation, and final submission
6	Jan 12	Final Report & Video Presentation

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